

IN AN EMERGENCY

111

the only number a student must know

A **LAN-only campus emergency & internal calling system** on Asterisk/FreePBX — runs over campus Wi-Fi with **no internet, no SIM, no cloud**, so it keeps working when normal networks are down. Deployable as a fixed campus install **or** a self-powered disaster-response van with corner repeaters. Every 111 call is recorded from the first second.

WHAT IT IS
Asterisk PBX in a QEMU Ubuntu VM on a Windows laptop; Android phones (Linphone) register as SIP softphones over the LAN. One-command installer stands up the whole stack + autostart.

IDENTITY MODEL
SAP ID = extension = username. Responder roles are **positions (4xxx)** staffed by shift — never a personal account. Same person + different job = same ID, different context.

WHY LAN-ONLY
Survives internet / cellular / cloud / PSTN failure — the point. **111 is human-first**; the offline coach (102) is the automatic fallback when no human answers. AI never gates 111.

THE SYSTEM IN ONE PICTURE
Android (Linphone) on Wi-Fi → AP → Switch/Router → **Asterisk / FreePBX** → Local storage (recordings · VM · CDR/CEL · incidents)
dial 111 → ert_emergency_queue (press 1 → first-aid) → offline coach 102 if no answer point free
ERT Lead 4101 + backup (Sec 4300 · Med 4200 · Warden) alerted in the background, in parallel – no dead-air
✗ no Internet ✗ no Cellular ✗ no Cloud ✗ no PSTN – excluded by design

EMERGENCY & SERVICE CODES
111 Campus emergency hotline — human-first, rings ERT queue, recorded
102 Offline panic-coach — deterministic first-aid, zero internet; auto fallback
101 Local-first AI triage (no cloud) — always escalates to 111; never dialed
199 Drill / test (no real dispatch) · **198** echo · **196** AI test
~~100~~ Deprecated & removed (softphones hijack it as "Police")

FEATURE CODES
***22 / *23** Responder shift on / off (join / leave 111 queue)
***45 / *46** Pause / resume within a shift
***77** Silent SOS / duress → pages ERT zone (everyone)
***9 · *80<ext>** Incident bridge → 9000 · intercom auto-answer (ERT/admin)
105 · 411 Request callback · dial-by-name directory
211-215 Dept hunt: Security / Medical / Warden / Ops / IT
720-723 Announce: evacuate / avoid / all-clear / assemble (ERT-Lead)

RESERVED RANGES & ZONES

SAP ID	
4101 / 4110-19 / 4120	ERT Lead · ERT operators (answer queue) · Control Room
4200/4300/4400/4500/4600s	Medical · Security (4301 Lead) · Warden · Admin/Ops · IT — dispatch targets
4700s	Fixed devices — speakers, gate phones
700-799	Paging zones (700 all-campus = PIN); logged
9000-9099	Incident-command conf rooms (9000 recorded, PIN)

INSIDE THE PBX · DEPLOYMENT
Components: chan_pjsip (SIP) · dialplan/contexts (routing + least-privilege) · app_queue · ert_emergency_queue · MixMonitor (recording) · voicemail · confbridge (9000s) · app_page (700s) · CDR/CEL · **upes_api** FastAPI · :8090 .
Modes: A — campus server (mains+UPS); B — self-powered van + corner repeaters (disaster / failover). Same config & numbers. HA option: 2× Jetson Nano active/standby (keepalived VIP).

ALSO IN THE BOX

- Operations Console** :8080 + two always-on **LED-TV wallboards** (safety / ops)
- Shared CardDAV phonebook** — dial responders by name · stable name upes-ecs.local (mDNS)
- Multi-language prompts** (Piper TTS): English + Hindi, Telugu, Malayalam, Urdu, Nepali · **missed-emergency follow-ups**

THE 111 CALL PATH — NO DEAD-AIR, NO SERIAL RINGING

Caller dials **111** • recording starts • incident ID assigned

ert_emergency_queue — ring ALL available ERT positions, 20s
press **1** any time —> first-aid fast-path (102)

answered —> **ERT: classify · dispatch · log · recording linked**

no answer point free —> IN PARALLEL:

Offline Panic-Coach 102
deterministic first-aid, zero internet. 9=retry 8=message
message
Emergency Voicemail (60s)

BACKGROUND alert (call-files):
Lead 4101 + Security 4300 + Medical 4200 + Warden
"press 1 to join the queue"

Missed Emergency Incident — Critical, Pending Review
(final catch — nothing is ever dropped)

RESPONDER POSITIONS — STAFFED BY SHIFT

EXT	ROLE	111 QUEUE?
4101	ERT Lead / Incident Cmdr	Escalation
4110-19	ERT Operators	ANSWERS
4120	Control Room	ANSWERS
4200s	Medical (4200 dispatch)	Dispatch
4300s	Security (4301 Lead)	Dispatch
4400/4500/4600s	Warden · Admin/Ops · IT	Dispatch
4700s	Fixed devices	No

ERT positions answer the queue; the rest are dispatch targets. Each dept = dispatch front-door + minimal seats.

GOLDEN RULES

- **111 is human-first** — never depends on AI / internet / cellular.
- **Unsure? Escalate.** Never drop an emergency call silently.
- **Only 111 / 199 are recorded** — student-to-student calls never are.
- **Emergency controls role-restricted · back up before every change; test 199 before 111.**

DONE & VALIDATED LIVE

- SIP reg on real Androids; two-way audio
- **111 rings a real handset** (on-shift)
- Offline coach 102; real 111 recording
- Status API :8090; VM autostart + 1-cmd deploy
- 32 SOPs · 8 Blueprints · Journal

BUILDABLE NEXT

- Real recorded voice prompts
 - QR / profile phone provisioning
 - Console served over LAN
 - Filled test-evidence sheet
- Deferred by choice:** DPDP compliance · cost/BOM.

NEEDS HARDWARE / DECISION

- **#1 — off QEMU NAT — real LAN IP** (van Linux / bridged)
- TLS + SRTP encryption
- Always-on ERT Androids · van UPS
- mDNS/DHCP · full pilot + go-live

SET UP A NEW WINDOWS PC — ONE COMMAND

```
powershell -ExecutionPolicy Bypass -File .\Install-UpesEcs.ps1
```

Prereqs + PBX VM + all accounts + autostart + Console + TV boards. Variants: **-LanIp** · **-LaunchTV** · **-Rebuild** · **-Uninstall**. Internet needed only for the first build.

ADD A PERSON — NEVER CHANGES AN EXISTING PASSWORD

```
powershell -File deploy\qemu\Add-UpesUser.ps1 -SapId 500123394 -Name "Kinjal Srivastava"
```

SAP ID = ext = username. Generates the secret once, pins it as single source of truth, pushes live, syncs roster + wallboard directory.

CAMPUS SCREENS & DAY-TO-DAY

Public safety board (giant DIAL 111)	:8080/tv-safety.html
Operations board (queue, KPIs)	:8080/tv-ops.html
Launch both TVs · re-point LAN IP	Show-TV -Both · Set-UpesLanIp

STORAGE & RETENTION

Recordings & voicemail	90 days
Incidents · CDR/CEL · queue/paging/conf logs	1 year
Config backups	30 daily + 12 weekly

No captured call audio is ever left on a server or shipped in an installer — only generated TTS prompts.

Remaining work is execution, not design.
Stand up on a real LAN IP → run the Pilot Test Plan → sign off go-live.

EMERGENCY
111